

Supplementary Material

It is Not the Standard, It is the Feed: Analysis of a Minimum Viable
Denominator for Comparing GTFS and NeTEx in European Public Transport Data

Livia Mihaela Miclaus, Shahrom Sohi
Vienna University of Economics and Business,
Institute of Data, Process and Knowledge Management, Vienna, Austria

This document is the supplementary material referenced in the main paper. It contains the extended methodological detail, the per-country matching procedures, and the full set of proportional overlap diagrams that were removed from the manuscript during revision in order to keep the paper within its page limit. Nothing here supersedes the paper: where a figure or value appears in both, the paper is authoritative.

The supplement is hosted alongside the analysis code and the underlying data at <https://github.com/DataFan2/gtfs-netex-mvd-comparison>.

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1 How this document relates to the paper

The paper reports results *per country*. This supplement retains the original presentation, in which countries were additionally grouped into five regional blocks: DACH (Austria, Germany, Switzerland), Western Europe (France, Belgium, Luxembourg, the Netherlands), Northern Europe

(Finland, Norway, Sweden), Southern Europe (Italy, Spain) and Central and Eastern Europe (the Czech Republic, Poland, Romania).

Note on the regional groupings. The regional labels in Figures S6–S12 are *presentational only*: they organise the per-country results and are not averages over complete regions. NeTeX was unavailable or unusable in several countries, so ten countries contribute to the stop comparison, nine to routes and six to calendars. Southern Europe and Central and Eastern Europe rest on one country each, Italy and the Czech Republic, and should be read as individual cases.

2 Dataset provenance in full

Table 1 of the paper summarises the source, publisher, scope, declared validity, access date and exclusions for each feed. Table S1 below records the remaining provenance detail that the paper’s Table 1 caption defers to this repository: the complete access URL, the exact file name of the artefact analysed, and the licence under which it was published. The same information is maintained in machine-readable form as `DATA_SOURCES.md` in the repository, alongside the per-country notebooks that consume each file.

Raw datasets are not redistributed in the repository. To reproduce the analysis, the files listed below must be downloaded from their source and placed in the corresponding country’s `data/raw/` folder; each notebook regenerates its own `data/processed/` outputs when run top to bottom.

Table S1. Full provenance record for each analysed feed. **G** = GTFS, **N** = NeTEx. Access dates are as reported in Table 1 of the paper. “Registration” marks sources that require an account, API key or signed licence before download.

Country	Fmt	Publisher	Access URL	File name analysed	Licence	Accessed
<i>Both formats available and compared</i>						
Austria	G	ÖBB (operator)	https://data.oebb.at/de/datensaetze/~soll-fahrplan-gtfs/	GTFS_Fahrplan_2026.zip	Not stated	2026-03-04
	N	Mobilitätsverbünde Austria (regional associations)	https://data.mobilitaetsverbuende.at <i>Registration required; no deep link available without an account</i>	netex_evu_2026.zip	Not stated	2026-03-05
Switzerland	G	Swiss Federal Office of Transport (BAV), NAP	https://data.opentransportdata.swiss/dataset/timetable-2026-gtfs2020/resource/1d8312b8-7e27-4cc6-b6ac-3207fa92b5b4	switzerland_gtfs_2026-04-01.zip	Non-commercial allowed / commercial allowed / reference required	2026-04-08
	N	Swiss Federal Office of Transport (BAV), NAP	https://data.opentransportdata.swiss/dataset/timetablenetex_2026/resource/cce2bb3c-2221-4f39-ab12-891e6b8dbf1e	switzerland_netex_2026-03-31.zip	Non-commercial allowed / commercial allowed / reference required	2026-04-08
France	G	SNCF (operator), via the French NAP	https://transport.data.gouv.fr/datasets/horaires-sncf <i>Downloaded from</i> https://ressources.data.sncf.com	Export_OpenData_SNCF_GTFS_NewTripId.zip	Not stated	2026-04-27
	N	SNCF (operator), via the French NAP	https://transport.data.gouv.fr/datasets/horaires-sncf <i>Downloaded from</i> https://ressources.data.sncf.com	export-opendata-sncf-netex.zip	Not stated	2026-04-27
Luxembourg	G	Verkéiersverbond (authority), via data.public.lu	https://data.public.lu/en/datasets/horaires-et-arrets-des-transport-publics-gtfs/	gtfs-20260408-20260430.zip	CC BY 4.0	2026-04-22
	N	Verkéiersverbond (authority), via data.public.lu	https://data.public.lu/en/datasets/horaires-et-arrets-des-transport-publics-netex/	netex-20260408-20260430.zip	CC BY 4.0	2026-04-22
Netherlands	G	DOVA (authority); distributed via OVapi	https://gtfs.ovapi.nl/nl/	NL-20260505.gtfs.zip (served as gtfs-nl.zip)	CC0 1.0	2026-05-05

Country	Fmt	Publisher	Access URL	File name analysed	Licence	Accessed
Romania	G	Community conversion of CFR Călători data published on data.gov.ro	https://github.com/vasile/data.gov.ro-gtfs-exporter/tree/master/gtfs-out	data.gov.ro-gtfs-exporter-master.zip	MIT	2026-05-18
<i>Excluded:</i> no NeTeX published. The GTFS feed is itself an unofficial community conversion rather than a direct CFR Călători publication.						

3 The Minimum Viable Denominator: extended rationale

The paper defines the MVD as the smallest set of concepts represented in both GTFS and NeTEx that together suffice to compare published datasets in practice, covering stop/station information, route/line information and calendar information. Three reasons motivated the selection of exactly these three dimensions.

First, they are legally anchored. The three dimensions match the minimum data categories required by *Level of Service 1* of EU Delegated Regulation 2017/1926. The MVD is therefore not an arbitrary choice: it targets the same minimum data that EU-wide travel information services are already legally required to provide.

Second, they are functionally minimal. They represent the three questions any journey planner must answer before any other: where can a passenger board, where is the vehicle going, and when does it run. Features beyond these three, such as fares or accessibility details, provide limited practical value without this baseline first being in place.

Third, each has a documented NeTEx equivalent. Stops correspond to `StopPlace` and `ScheduledStopPoint`; routes correspond to `Line`. Calendar information is the exception, and the reason it is treated separately in Section 5.3: the Data4PT mapping splits it across several linked elements rather than a single equivalent, with `calendar.txt` corresponding to `PropertyOfDay` and `OperatingPeriod`, and `calendar_dates.txt` corresponding to `DayType` and `DayTypeAssignment`.

A direct one-to-one comparison between the two standards cannot be assumed, because GTFS groups related information into flat tables while NeTEx uses a modular XML schema organised into frames, each covering a different functional area. The purpose of the MVD is not to capture everything either standard can express, but to reduce both formats to simplified, comparable tables covering the three core dimensions above.

4 Country selection and data search

The fifteen countries were selected to represent a broad and diverse set of European regions and data-publishing practices. For each country, GTFS and NeTEx datasets were searched independently, prioritising official sources such as National Access Points, governmental open-data portals, railway operator portals and transport-specific national platforms; third-party or community-maintained sources were used only where no official source could be found.

Access conditions varied considerably. Some datasets were openly downloadable, while others required registration, an API key, or licence approval, and some covered only a region rather than the whole country. GTFS was generally easier to access than NeTEx: a usable GTFS dataset was found for all fifteen countries, while no usable NeTEx dataset could be obtained for four of them (Belgium, Poland, Romania and Spain). Because a country-level MVD comparison requires both formats to be available and processable, these four countries, together with cases where NeTEx was too fragmented or structurally inconsistent to support a full comparison (Germany, and Finland for the calendar dimension), could not be carried through to all of the results. This is why several regional blocks in Section 7 include fewer countries than the blocks defined in Section 1.

5 Matching procedures

The matching technique had to vary from country to country, because the underlying GTFS and NeTEx data was not published in the same way everywhere. The fields, labels and structures available for each dimension differed in completeness and reliability from one country to the next,

so the technique used had to adapt to what the data actually offered rather than follow one fixed approach.

5.1 Stop and station matching

The stop/station comparison is based on the GTFS `stops.txt` fields `stop_id`, `stop_name`, `stop_lat` and `stop_lon`, compared against the NeTEx `StopPlace` element. Figure S1 shows how these fields map onto `StopPlace`: `stop_id` corresponds to the `id` attribute, `stop_name` to the `Name` child element, and `stop_lat`/`stop_lon` to `Latitude`/`Longitude`, nested two levels deeper inside `Centroid` and `Location`.

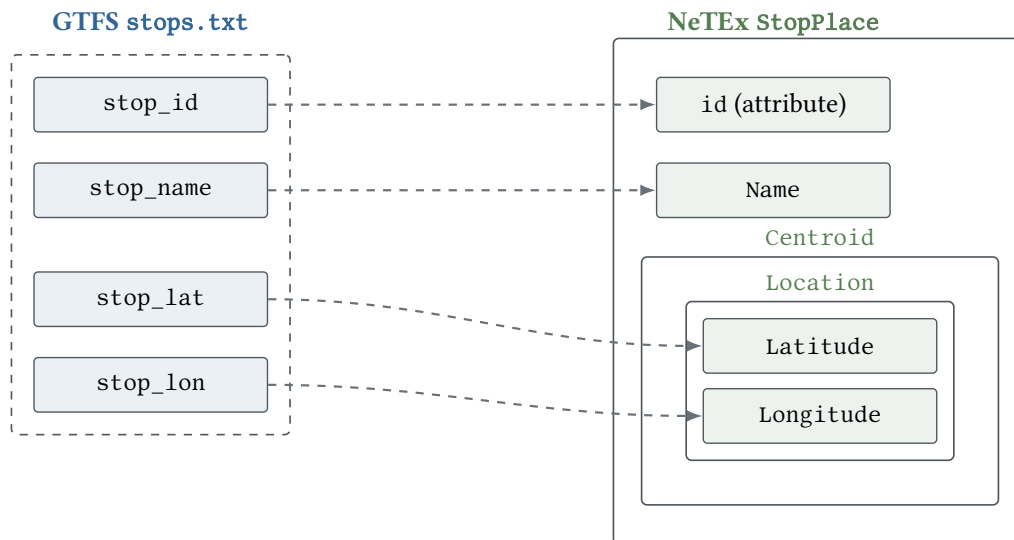


Figure S1. Mapping from flat GTFS `stops.txt` fields (left) to the nested NeTEx `StopPlace` element (right). Coordinates in NeTEx are reached through `Centroid` → `Location` → `Latitude/Longitude`, rather than a flat column as in GTFS. (Figure 1 in the original manuscript.)

The technique used to link GTFS and NeTEx records on this basis varied from country to country, because the two datasets did not offer the same starting point everywhere. Where GTFS and NeTEx identifiers were shared, or could be made comparable through simple cleaning, matching relied on identifiers. Where identifiers were missing, incompatible, or could not be independently verified, matching relied instead on normalized names, geographic coordinates, or both. Table S2 summarises the technique used per country.

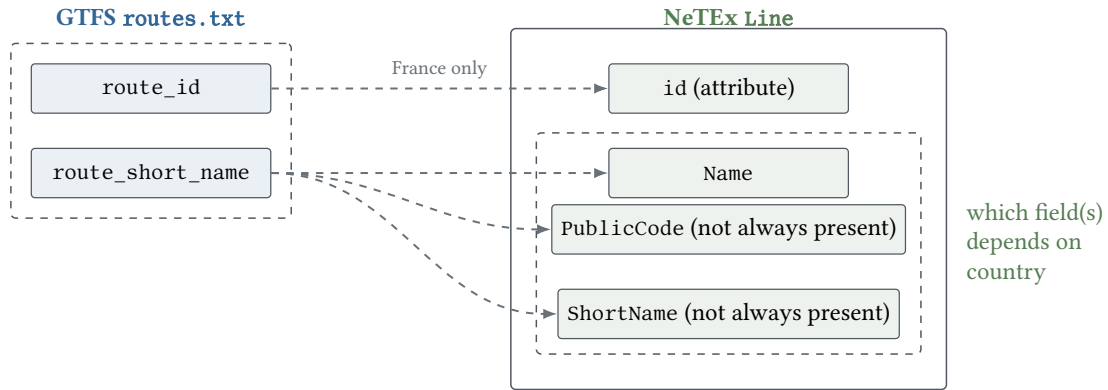
5.2 Route and line matching

The route/line comparison is based on two GTFS `routes.txt` fields: `route_id`, a technical identifier not meant for passengers (for example 1-MB4-j26-1), and `route_short_name`, the public-facing line label passengers actually see (for example S4). These were compared against the NeTEx `Line` element. Figure S2 shows how the fields map: `route_id` corresponds to the `id` attribute (for example `at:obb:Line:01MB4_.j26:`), while `route_short_name` was compared against whichever of `Name`, `PublicCode` or `ShortName` was actually populated on the NeTEx side, since all three can hold the same kind of short public label (for example `PublicCode` = "S4"), but not every NeTEx `Line` element fills in all three.

The way records were actually linked also differed by country, not just the field used. Where GTFS `route_id` and NeTEx `Line id` were shared, the two tables were joined directly, row by row, on this identifier (France only). Everywhere else, a row-level join was not meaningful. This is because GTFS and NeTEx each create a separate technical record for every direction, operator, or journey-pattern variant of a line, even though a passenger ever sees only one line label. In

Table S2. Stop/station matching strategy used per country. (*Table 1 in the original manuscript.*)

Category	Technique	Countries
Identifier-based, confirmed	Shared or convertible stop identifiers, matched directly or after cleaning	Switzerland, Norway, Luxembourg
Identifier-based, unconfirmed	Numeric identifiers overlapped, but with no coordinates or names available to verify that the overlap reflected the same physical stop	Czech Republic
Name-based, coordinate-validated	Stop names normalized and matched exactly, with a coordinate distance check used to confirm each match	Austria
Coordinate-based (nearest-neighbour)	Stops matched to the nearest stop in the other format within a fixed distance threshold	France, Finland, Italy, Sweden, Netherlands
Not comparable	No stop/station comparison could be performed because NeTEx data were unavailable	Germany, Belgium, Poland, Romania, Spain

**Figure S2.** Mapping from GTFS `routes.txt` fields to the NeTEx `Line` element used for route/line matching. (*Figure 2 in the original manuscript.*)

Switzerland, for example, the public label `PB` existed as 54 separate GTFS route rows and 44 separate NeTEx line rows, all representing the same real-world line. Joining on raw rows would have produced many false mismatches simply due to this technical splitting, not because the lines were actually different. Instead, each side was first reduced to a set of unique, deduplicated public labels, and these two label sets were compared for overlap: a label counted as matched if it appeared in both sets, regardless of how many underlying technical records it corresponded to on either side.

For two countries, the available fields did not support a methodologically valid matching procedure. Italy’s GTFS feed itself only covers the Tuscany region, while its NeTEx file covers the entire national Trenitalia network. However, the mismatch was not primarily attributable to differences in geographic scope. In Italy, GTFS `route_long_name` described a physical route (for example `Firenze - Pisa - Livorno`), while NeTEx `PublicCode` described a service category (for example `REG`, for a regional train). These are two different dimensions of the same network, not two labels for the same object, so no amount of cleaning could align them, and this would have remained true even with matching geographic scope on both sides. In the Czech Republic, GTFS `route_short_name` held a public train name (for example `Ex32 Baltic express`), while NeTEx `PublicCode` held an internal numeric train number (for example `1028`). Both describe the same kind of thing, a specific train service, but using naming conventions that do not overlap as text, and no reference table linking the two was available in the published data.

Table S3. Route/line matching technique used per country. (*Table 2 in the original manuscript.*)

Category	Technique	Countries
Identifier-based	Shared line identifier, confirmed by a direct join between GTFS <code>route_id</code> and NeTEx <code>Line id</code>	France
Label-based, normalized	GTFS <code>route_short_name</code> matched against NeTEx <code>PublicCode</code> or <code>Name</code> , with only light formatting cleanup (case, whitespace, hyphen suffixes)	Switzerland, Norway, Luxembourg, Austria
Label-based, fallback-patched	GTFS <code>route_short_name</code> matched against NeTEx <code>PublicCode</code> , with <code>route_long_name</code> substituted in where <code>route_short_name</code> was missing, lowering match quality	Finland, Sweden
Structurally incompatible labeling	GTFS and NeTEx labeled lines using fundamentally different systems, so label matching produced no usable result regardless of data quality	Italy, Czech Republic
Not comparable	No country-wide route/line comparison could be produced because NeTEx data were unavailable	Germany, Belgium, Poland, Romania, Spain, Netherlands

5.3 Calendar matching

Unlike the stop/station and route/line dimensions, the calendar comparison did not require a different matching technique for each country. Once each side had been reduced to a comparable set of active service dates, the same rule, an exact day-by-day comparison of those dates, was applied to every country. What still varied by country was not this final rule, but the path taken to reach it. As already noted, Data4PT does not map GTFS `calendar.txt` and `calendar_dates.txt` onto a single NeTEx element. The equivalent information is split across `PropertyOfDay`, `OperatingPeriod`, `DayType` and `DayTypeAssignment`. Because these elements can be combined, referenced and populated differently from one publisher to the next, each country still needed its own extraction path to reach a comparable set of active dates.

Figures S3 and S4 show the NeTEx element chain used to reconstruct active service dates for each country, split into two figures by which GTFS calendar files were available as the starting point. Figure S3 covers Austria, Luxembourg, Norway and Sweden, where GTFS feeds published both `calendar.txt` and `calendar_dates.txt`. Figure S4 covers France and Italy, where GTFS feeds published only `calendar_dates.txt`, listing individual service dates directly rather than a weekly pattern with exceptions. The split is based on which GTFS files each country published, not on any difference in NeTEx.

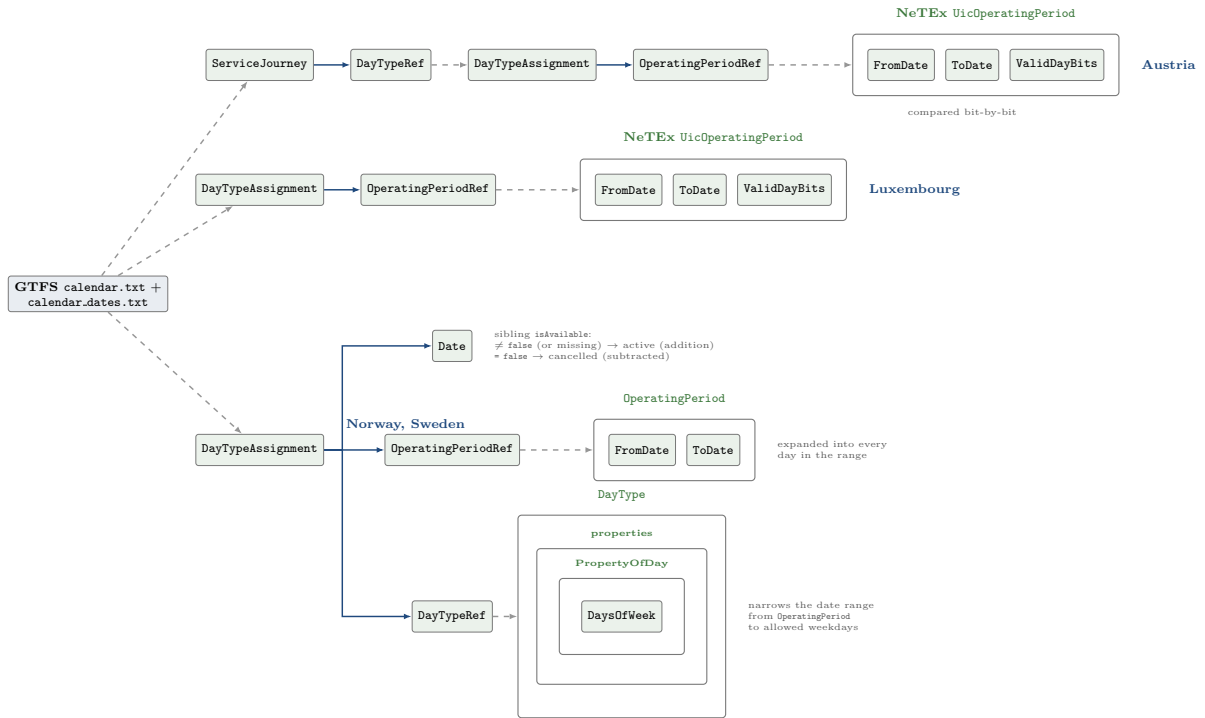


Figure S3. GTFS `calendar.txt` + `calendar_dates.txt` source: NeTEx element chain used for calendar comparison in Austria, Luxembourg, Norway and Sweden. (*Figure 3 in the original manuscript.*)

How GTFS represents calendar information

GTFS represents calendar information in two files. `calendar.txt` lists weekly patterns: a `service_id`, one flag per weekday, and a `start_date`/`end_date` validity window. For example, one row in Austria's GTFS feed reads

```
"TA+03100","0","0","0","0","0","1","1","20251214","20261212"
```

a pattern that runs only on Saturdays and Sundays between 14 December 2025 and 12 December 2026. `calendar_dates.txt` then adds or removes individual dates from that pattern, using an

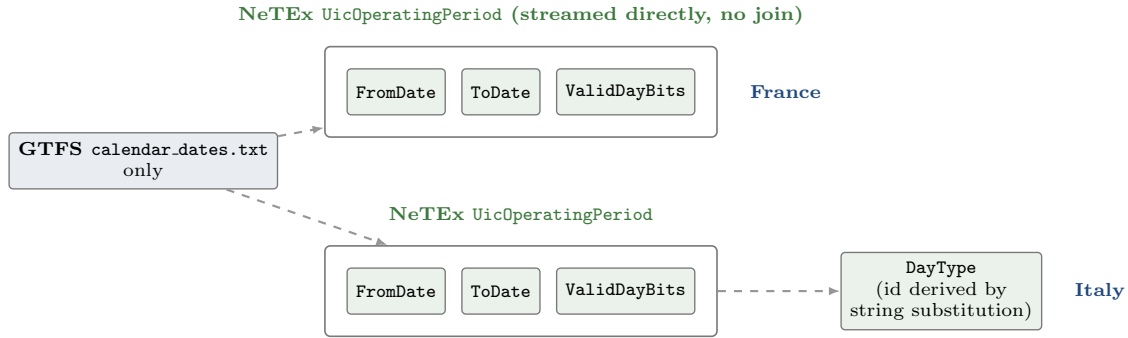


Figure S4. GTFS `calendar_dates.txt`-only source: NeTEx element chain used for calendar comparison in France and Italy. (*Figure 4 in the original manuscript.*)

exception_type of 1 (added) or 2 (removed). For this same service, `calendar_dates.txt` contains the rows

"TA+03100","20251225","1", "TA+03100","20251226","1", "TA+03100","20260101","1"

adding 25 December, 26 December and 1 January as served dates even though none of them falls on a Saturday or Sunday: this normally weekend-only service is extended to also run on Christmas Day, St. Stephen's Day and New Year's Day.

How NeTEx represents the same information

NeTEx does not hold this same information in a single element. Instead, a day category, its weekly pattern, its validity window and its date-level exceptions are each represented and linked together, as shown in Figures S3 and S4. Table S4 defines each one on its own.

Table S4. NeTEx elements used in calendar comparison, with real examples drawn from the source data. (*Table 3 in the original manuscript.*)

NeTEx element	Definition	Example
DayType	A named category of day that trips can be linked to, comparable to a GTFS <code>service_id</code>	<code>id="AKT:DayType:119_1"</code>
PropertyOfDay / DaysOfWeek (nested inside DayType)	States which days of the week the DayType covers, for example <code>Weekdays</code> or <code>Saturday</code>	<code><DaysOfWeek>Weekdays</DaysOfWeek></code>
DayTypeAssignment	Links a DayType to either a date range (OperatingPeriodRef) or one specific date (Date); <code>isAvailable</code> states whether that date is served (<code>true</code>) or excluded (<code>false</code>)	<code>Date: 2026-05-14</code> <code>DayTypeRef: DayType:100_1</code> <code>isAvailable: false</code>
OperatingPeriod	A date range only, with no day-by-day detail, just a first and last possible day	<code>FromDate: 2026-08-12</code> <code>ToDate: 2026-09-28</code>
UicOperatingPeriod	An extended date range that adds <code>ValidDayBits</code> , a string with one character per day marking it active (1) or inactive (0)	<code>FromDate: 2025-12-14</code> <code>ToDate: 2026-12-12</code> <code>ValidDayBits:</code> <code>100000110001111...</code>

The comparison rule

Once the relevant data has been extracted for each id, following the chains shown in Figures S3 and S4, the two sides are compared using one fixed method, applied the same way for every country. Figure S5 shows this method schematically.

A simplified example helps illustrate how it works. Suppose a GTFS service is active on 1, 3 and 4 December, and a NeTEx pattern is active on 1, 3, 4 and 6 December. If GTFS's own data only covers up to 5 December, the shared comparison window is 1 through 5 December, and NeTEx's 6 December is dropped, since it falls outside what GTFS covers at all. Each side's active dates within that window are then written as a string of 1s and 0s, one character per day: a 1 for an active day, a 0 for an inactive one. GTFS becomes 10110, and NeTEx, once 6 December is cut off, becomes the same string. Two patterns count as a match only if every character in this string is identical.

For clarity, this example uses simplified ids and dates rather than the real field names. In practice, the GTFS side of this comparison is built from `service_id` together with the `calendar.txt` and `calendar_dates.txt` fields described above, while the NeTEx side is built from the `DayTypeAssignment`, `OperatingPeriod`, `UicOperatingPeriod` and `DaysOfWeek` elements defined in Table S4.

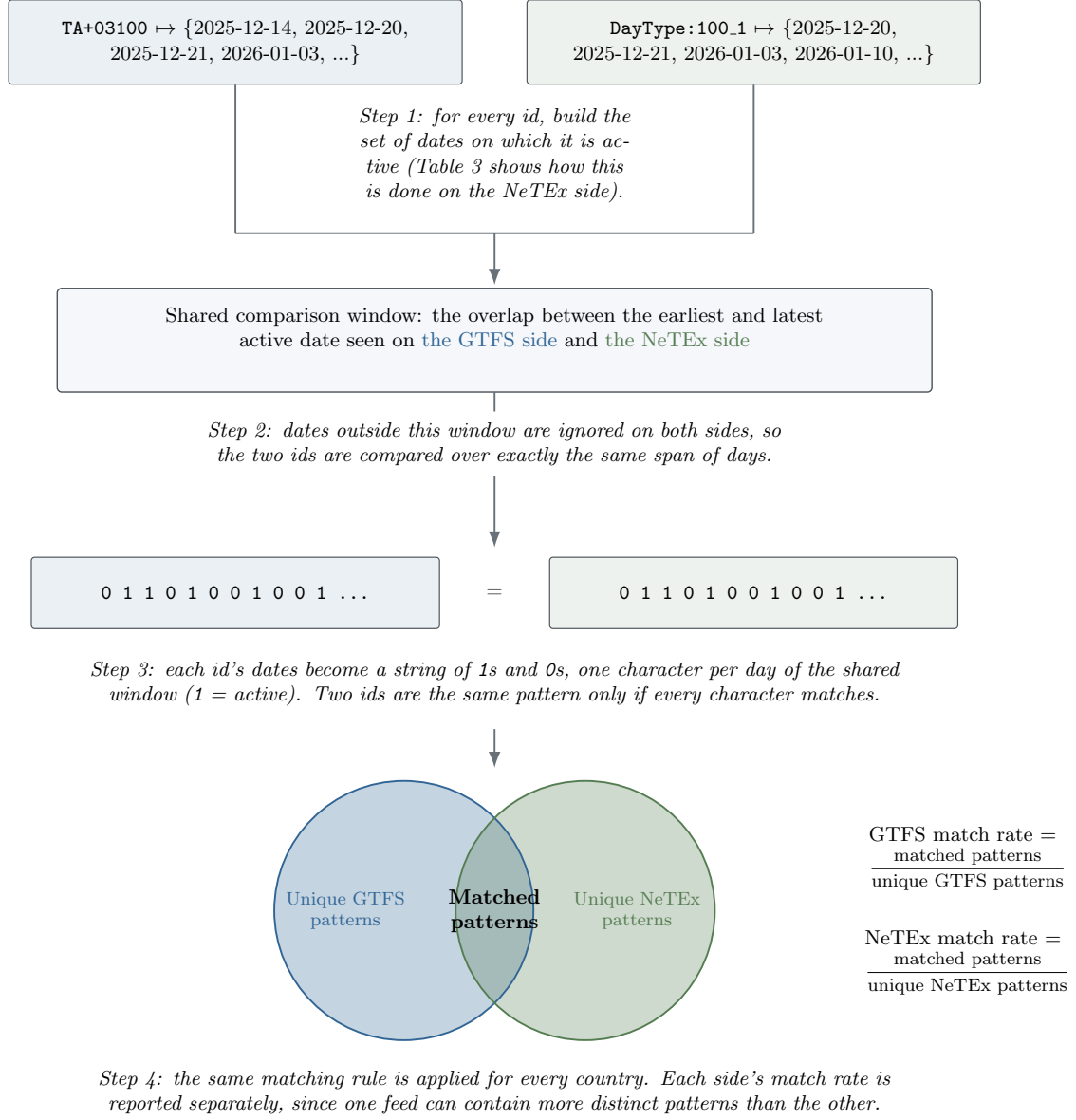


Figure S5. The calendar matching method, applied identically to every country: active dates are restricted to a shared comparison window, encoded as day-by-day bit strings, and compared as sets of unique patterns. (Figure 5 in the original manuscript.)

6 Interpretation of match rates

Each country is reported with two match rates, one for the GTFS side and one for the NeTEx side, each judged separately against the thresholds in Table S5. These thresholds are interpretive conventions rather than derived values, and are reproduced here for reference; the paper reports the underlying percentages directly.

Table S5. Thresholds used to interpret GTFS and NeTEx match rates. (*Table 4 in the original manuscript.*)

Match rate	Category	Interpretation
Below 50%	Weak match	Fewer than half of that side’s patterns have a counterpart on the other side.
50%–80%	Good match	More than half, but not the large majority, of that side’s patterns have a counterpart on the other side.
Above 80%	Excellent match	Nearly all of that side’s patterns have a counterpart on the other side.

7 Proportional overlap diagrams

Circle areas are proportional to each side’s set size and the overlap to the matched units; both coverage rates and the absolute counts are given beneath each diagram.

7.1 Stops and stations

Stop and station overlap, for the ten countries where both feeds could be compared.

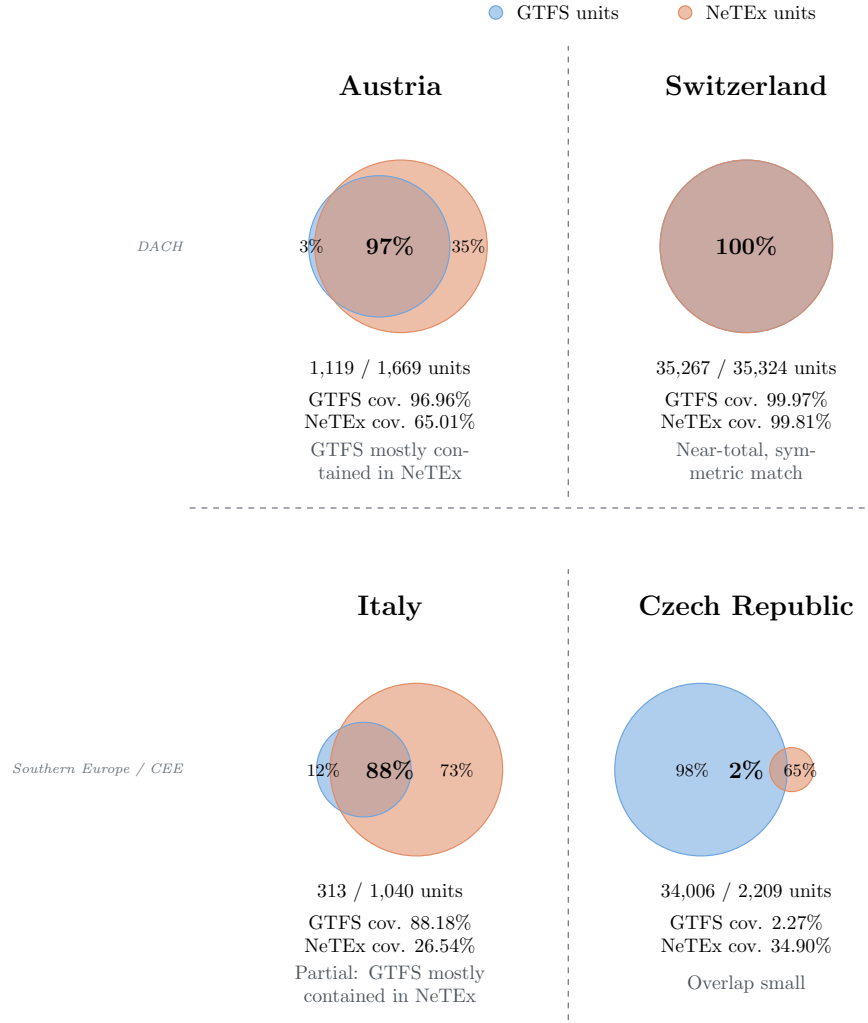


Figure S6. Proportional overlap of GTFS and NeTEx stop/station units: Austria and Switzerland (top), Italy and the Czech Republic (bottom). (*Figure 6 in the original manuscript.*)

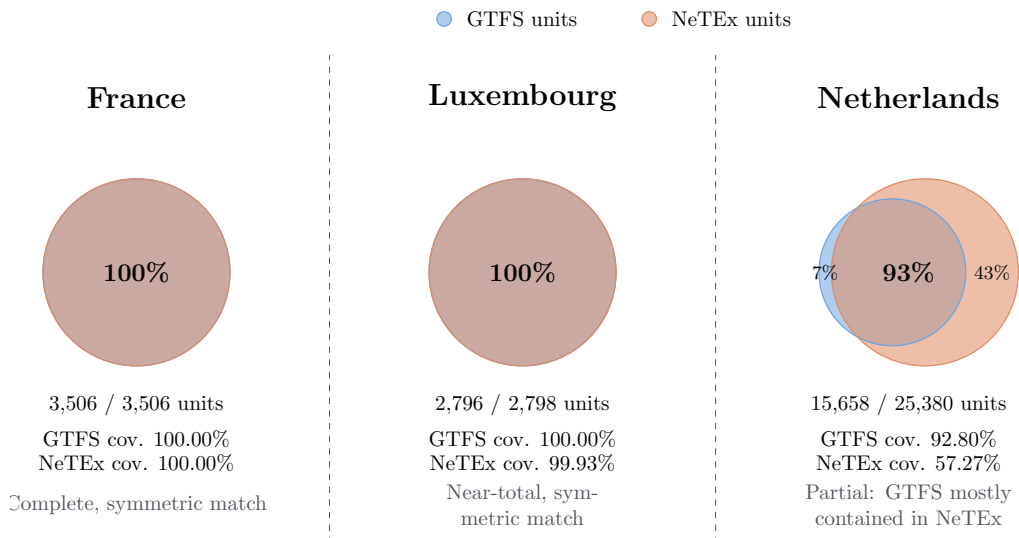


Figure S7. Proportional overlap of GTFS and NeTEx stop/station units: France, Luxembourg and the Netherlands. (*Figure 7 in the original manuscript.*)

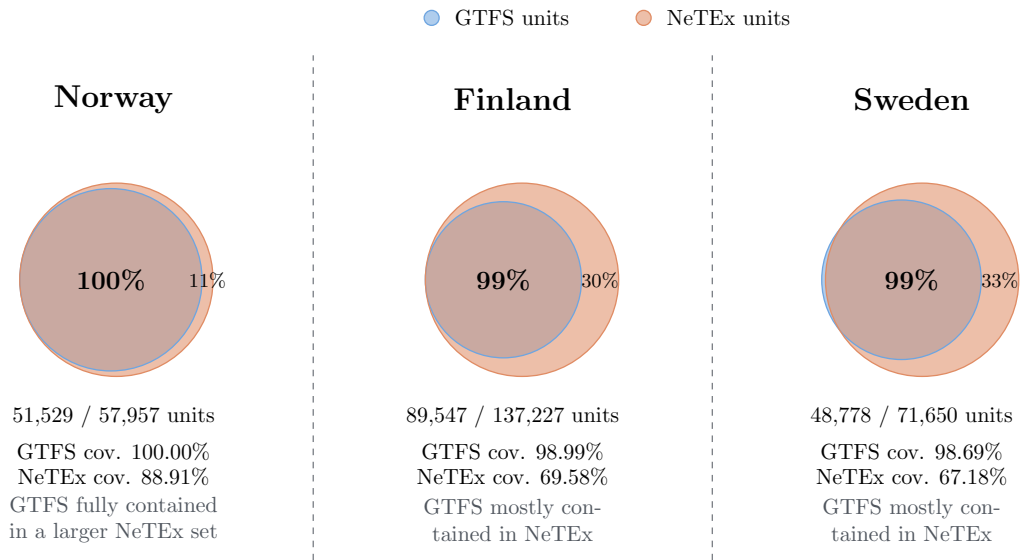


Figure S8. Proportional overlap of GTFS and NeTEx stop/station units: Norway, Finland and Sweden.
(Figure 8 in the original manuscript.)

7.2 Routes and lines

Route and line label overlap, for the nine countries where both feeds could be compared.

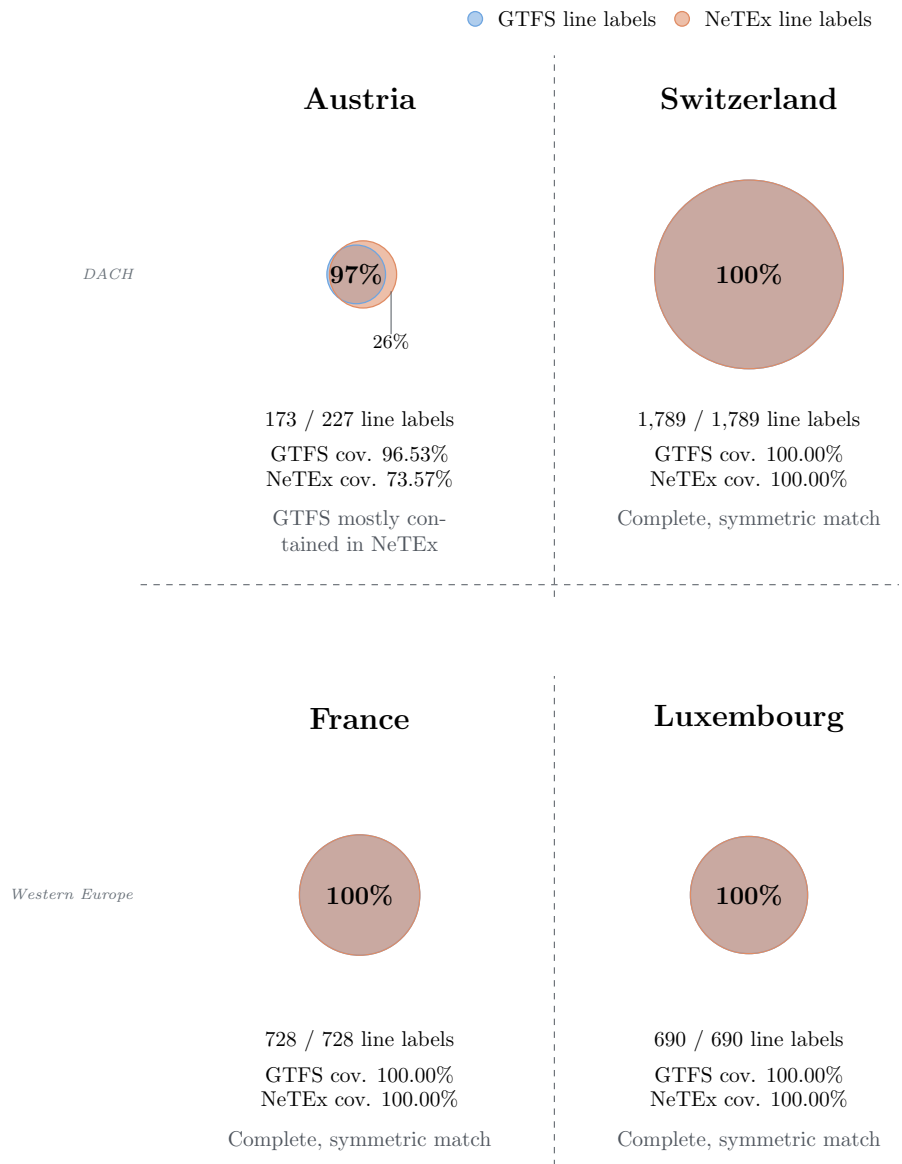


Figure S9. Proportional overlap of GTFS and NeTEx route/line labels: Austria and Switzerland (top), France and Luxembourg (bottom). (*Figure 9 in the original manuscript.*)

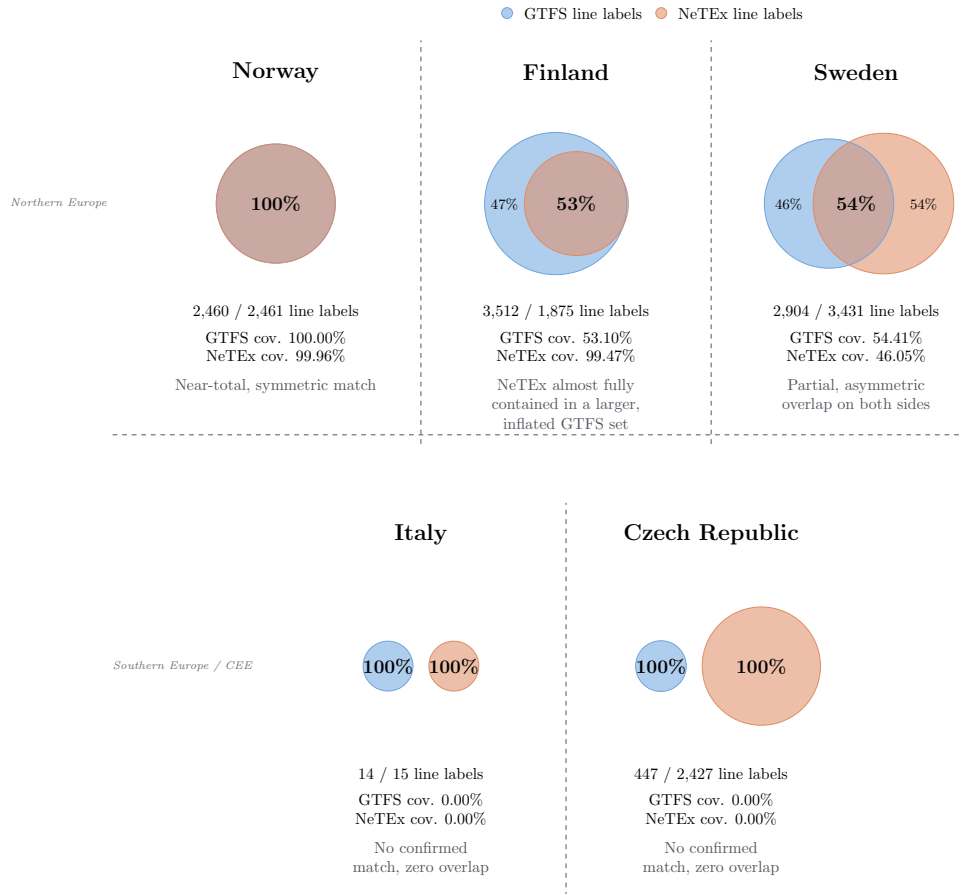


Figure S10. Proportional overlap of GTFS and NeTEx route/line labels: Norway, Finland and Sweden (top), Italy and the Czech Republic (bottom). (*Figure 10 in the original manuscript.*)

7.3 Service calendars

Active-day pattern overlap, for the six countries where both feeds could be compared.

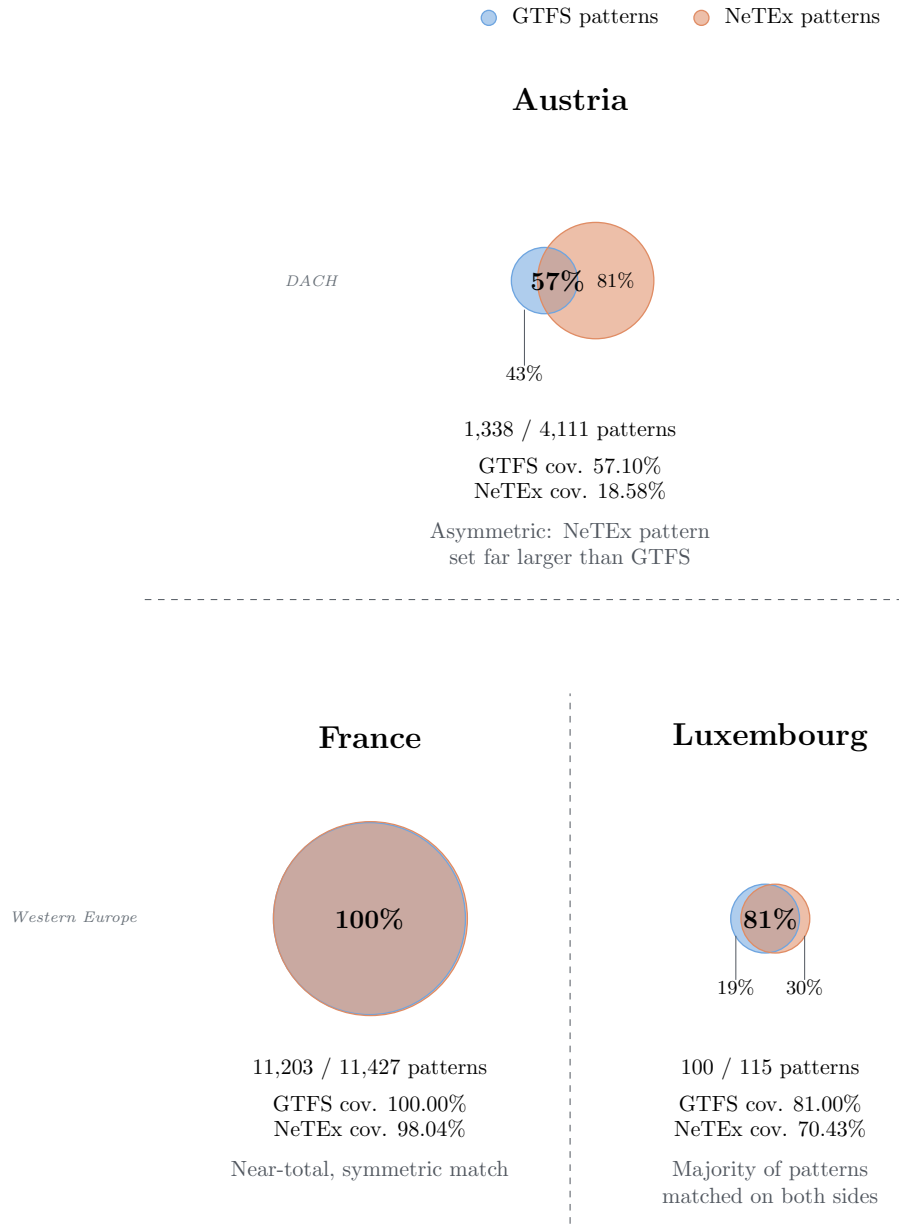


Figure S11. Proportional overlap of GTFS and NeTEx unique active-day patterns: Austria (top), France and Luxembourg (bottom). (*Figure 11 in the original manuscript.*)

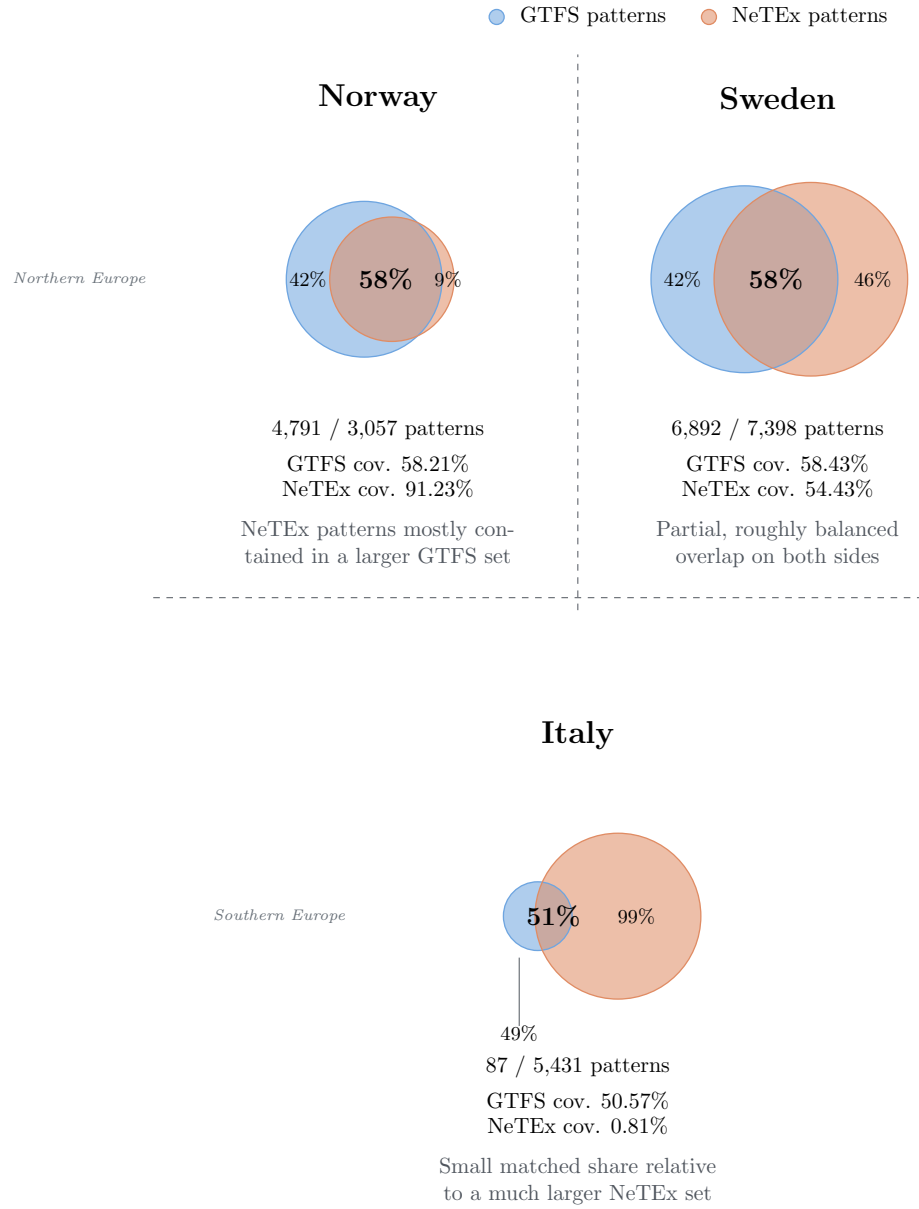


Figure S12. Proportional overlap of GTFS and NeTEx unique active-day patterns: Norway and Sweden (top), Italy (bottom). (*Figure 12 in the original manuscript.*)

8 Declaration on generative AI

During the preparation of this work, the author(s) used ChatGPT and Claude in order to: check grammar and spelling, assist with code debugging, extract and document methodological details from the analysis notebooks, and draft and revise passages of the manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take full responsibility for the publication's content.